WORKED SOLUTION

STEP III 2008, Question 7

Pure • Complex number

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q7

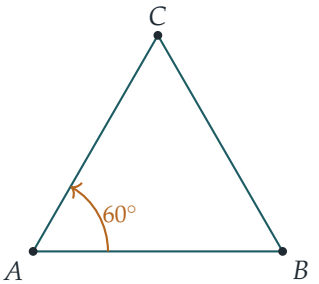
The key idea

In the Argand diagram, multiplying by $e^{i\pi/3}$ rotates through $60°$ anticlockwise. An equilateral triangle is exactly one side rotated through $60°$ about a vertex, so the opening relation turns "equilateral" into a linear equation in a, b, c. After that, both parts are bookkeeping: write every new point in terms of the old ones and check the condition we want, whether that is $p + r = q + s$ for a parallelogram or the opening relation itself for an equilateral triangle.

Solution

The opening relation

Write $\omega = e^{i\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$. Since ABC is equilateral, AC has the same length as AB and makes an angle of $60°$ with it. Because the triangle is described anticlockwise, we turn from $\overrightarrow{AB}$ to $\overrightarrow{AC}$ *anticlockwise*. So $\overrightarrow{AC}$ is $\overrightarrow{AB}$ rotated through $+60°$:



$$c - a = \omega(b - a) = \tfrac{1}{2}(b - a) + \tfrac{\sqrt{3}}{2}i(b - a),$$
$$2c = 2a + (b - a) + i\sqrt{3}(b - a) = (a + b) + i\sqrt{3}(b - a),$$

which is the required result.

If the triangle is described clockwise, $\overrightarrow{AC}$ is $\overrightarrow{AB}$ rotated through $-60°$, so $c-a=\bar{\omega}(b-a)$ with $\bar{\omega}=\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}$. The same algebra gives

$$2c=(a+b)-\mathrm{i}\sqrt{3}(b-a).$$

Conversely, any c satisfying either relation makes ABC equilateral, because each relation is equivalent to $c-a=\omega^{\pm 1}(b-a)$. So to *show* that a triangle is equilateral, it is enough to verify one of these relations.

Part (i)

We get to choose the four apexes, and each one may go on either side of its edge. Using the opening results with the apex in the role of c, put

$$\begin{aligned}
2p&=(d+e)+\mathrm{i}\sqrt{3}(e-d) && (D,E,P \text{ anticlockwise}),\\
2q&=(e+f)-\mathrm{i}\sqrt{3}(f-e) && (E,F,Q \text{ clockwise}),\\
2r&=(f+g)+\mathrm{i}\sqrt{3}(g-f) && (F,G,R \text{ anticlockwise}),\\
2s&=(g+d)-\mathrm{i}\sqrt{3}(d-g) && (G,D,S \text{ clockwise}).
\end{aligned}$$

By the opening work, each of PDE, QEF, RFG, SGD is then equilateral.

Ask yourself Why alternate the orientations, rather than putting every triangle on the same side of its edge?

Try it and see. $PQRS$ is a parallelogram exactly when $\overrightarrow{PQ}=\overrightarrow{SR}$, that is, when $q-p=r-s$, or $p+r=q+s$. (This says that the diagonals PR and QS have the same midpoint.) With the choices above,

$$\begin{aligned}
2(p+r)&=(d+e+f+g)+\mathrm{i}\sqrt{3}\,(e-d+g-f),\\
2(q+s)&=(d+e+f+g)-\mathrm{i}\sqrt{3}\,(f-e+d-g)=(d+e+f+g)+\mathrm{i}\sqrt{3}\,(e-d+g-f).
\end{aligned}$$

These are equal, so $p+r=q+s$ and $PQRS$ is a parallelogram. If instead we had used the "+" sign throughout, the imaginary parts would have been $\mathrm{i}\sqrt{3}(e-d+g-f)$ and $\mathrm{i}\sqrt{3}(f-e+d-g)$. These agree only when $d+f=e+g$, that is, only when $DEFG$ is already a parallelogram. The alternating signs are what make the $\sqrt{3}$ terms match for every quadrilateral.

(For some quadrilaterals the four points could happen to be collinear, giving a degenerate parallelogram. The condition $\overrightarrow{PQ}=\overrightarrow{SR}$ still holds in that case.)

Part (ii)

If LMN is described clockwise, reflect the whole figure in the real axis (that is, take complex conjugates). Reflection preserves equilateral triangles, centroids, and whether a triangle on a side is drawn externally, so it is enough to treat the case where LMN is described *anticlockwise*.

Then N is to the left of the directed edge $L \to M$, and the external triangle LMX on LM has its apex X to the right, so L, M, X is described clockwise:

$$2x = (l+m) - \mathrm{i}\sqrt{3}(m-l).$$

The centroid of LMX is

$$u = \tfrac{1}{3}(l+m+x) = \tfrac{1}{3}(l+m) + \tfrac{1}{6}(l+m) - \tfrac{\mathrm{i}\sqrt{3}}{6}(m-l) = \frac{l+m}{2} + \frac{\mathrm{i}(l-m)}{2\sqrt{3}},$$

using $\frac{\sqrt{3}}{6} = \frac{1}{2\sqrt{3}}$. Cycling $l \to m \to n \to l$, which keeps the anticlockwise order, gives

$$v = \frac{m+n}{2} + \frac{\mathrm{i}(m-n)}{2\sqrt{3}}, \qquad w = \frac{n+l}{2} + \frac{\mathrm{i}(n-l)}{2\sqrt{3}}.$$

We show that U, V, W satisfy the anticlockwise relation $2w = (u+v) + \mathrm{i}\sqrt{3}(v-u)$. First,

$$u + v = \tfrac{1}{2}l + m + \tfrac{1}{2}n + \frac{\mathrm{i}(l-n)}{2\sqrt{3}},$$

$$v - u = \tfrac{1}{2}(n-l) + \frac{\mathrm{i}(2m-l-n)}{2\sqrt{3}},$$

$$\mathrm{i}\sqrt{3}(v-u) = \frac{\mathrm{i}\sqrt{3}}{2}(n-l) - \tfrac{1}{2}(2m-l-n).$$

? **Ask yourself** How does the m disappear?

Add the first and third lines. The real parts give $\frac{1}{2}l + m + \frac{1}{2}n - m + \frac{1}{2}l + \frac{1}{2}n = l + n$, so m cancels from the real part, and it never appeared in the imaginary parts. The imaginary parts combine as

$$(n-l)\Big(\frac{\sqrt{3}}{2} - \frac{1}{2\sqrt{3}}\Big)\mathrm{i} = (n-l)\,\frac{3-1}{2\sqrt{3}}\,\mathrm{i} = \frac{\mathrm{i}(n-l)}{\sqrt{3}}.$$

Hence

$$(u+v) + \mathrm{i}\sqrt{3}(v-u) = (n+l) + \frac{\mathrm{i}(n-l)}{\sqrt{3}} = 2w.$$

By the opening result (and its converse), UVW is an equilateral triangle, described anticlockwise. This is Napoleon's theorem.

Checks

- Opening relation: $a = 0$, $b = 2$ gives $c = 1 + \mathrm{i}\sqrt{3}$, the apex above the real axis, as it should be for an anticlockwise triangle.
- Part (ii) with $l = 0$, $m = 6$, $n = 6\mathrm{i}$: the formulas give $u = 3 - \mathrm{i}\sqrt{3}$, $v = (3+\sqrt{3})(1+\mathrm{i})$ and $w = -\sqrt{3} + 3\mathrm{i}$. All three sides of UVW then have squared length $24 + 12\sqrt{3}$.
- We tested the opening relations, the parallelogram condition and Napoleon's theorem numerically on 10^4 random configurations. We also confirmed that using the same orientation on every edge in part (i) generally fails.

Common slips

- Getting the external direction wrong in part (ii). For an anticlockwise LMN the external apex on LM is *clockwise* from L, M. With the wrong sign we get the inner Napoleon triangle instead. (That one is also equilateral, but it isn't what the question asks for.)
- In part (i), trying to make all four triangles point the same way. That works only when $DEFG$ is itself a parallelogram.
- Testing for a parallelogram with $p + q = r + s$. Opposite vertices pair up: the condition is $p + r = q + s$.

Beyond STEP

The underlying idea is that the maps $z \mapsto \alpha z + \beta$ are exactly the rotations, enlargements and translations of the plane, so geometric conditions become linear equations over $\mathbb{C}$. A neat way to restate the opening relation is that ABC is equilateral precisely when $a + \omega^2 b + \omega^4 c = 0$ or $a + \omega^4 b + \omega^2 c = 0$, and Napoleon's theorem then drops out of a few lines of algebra. Needham's *Visual Complex Analysis* develops this geometric view of complex multiplication beautifully.

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